Francesca Giardine

giardine@illinois.edu | Google Scholar | Personal Website

Education

Ph.D. **University of California, Berkeley** May 2026
Doctor of Philosophy in Electrical Engineering and Computer Sciences
Dissertation: *"Analysis and Operation of Flying Capacitor Multilevel Converters: Single-Phase Applications and Fundamental Electromagnetic Interference Scaling"*
Advisor: Dr. Robert Pilawa-Podgurski

B.S. **Smith College** May 2020
Bachelor of Science in Engineering Science & Mathematics and Statistics
Summa Cum Laude

Honors and Awards

Fellowships

National Science Foundation Graduate Research Fellowship Sept. 2022 – Sept. 2025
- Tuition and stipend for 3 years of graduate study in STEM ($148,000 total).

Hearts to Humanity Eternal Graduate Research Grant Sept. 2023 – Sept. 2024
- $10,000 grant for independent research.

Berkeley EECS Department Fellowship Aug. 2020
- Semester-long fellowship for select incoming students.

Recognitions

Rising Stars in EECS Oct. 2025
- Selective 2-day program organized by MIT and Boston University to provide preparation for junior faculty positions.

WiscProf Future Faculty Program Participant May 2025
- Selective 4-day program sponsored by the University of Wisconsin, Madison providing training for faculty positions in academia.

Best Paper Award, ECCE Europe 2024 Sept. 2024
- Awarded to 10 of 300 papers presented at ECCE Europe for paper [C4].

NextProf Nexus Future Faculty Program Participant Aug. 2024
- Selective program sponsored by the University of Michigan, UC Berkeley, and Georgia Tech giving participants the opportunity to explore and prepare for a faculty position in academia.

Adeline Devor Penberthy, Class of 1945, Memorial Prize May 2020
- Awarded to an undergraduate engineering major for academic excellence in engineering and outstanding contributions toward building a community of learners within the Smith College Picker Engineering Program.

Research Experience

Postdoctoral Research Associate in the Stillwell Power Group Aug. 2026 – Present
University of Illinois Urbana-Champaign Urbana, IL
- Developing electromagnetic interference (EMI) test setup for the power research area, with an emphasis on long-term usability.
- Analyzing EMI in power converters for electrified transportation.
- Mentoring student research projects, specifically in EMI and isolated power converters.

Pilawa Research Group Aug. 2020 – May 2026
University of California, Berkeley Berkeley, CA

- Studied electromagnetic interference (EMI) for the flying capacitor multilevel converter.
- Formulated modulation technique to improve efficiency and EMI of flying capacitor multilevel inverters.
- Simulated and designed advanced hybrid switched-capacitor converters for use in DC-AC grid-connected systems.

Power Electronic Device Characterization Internship June 2023 – Aug. 2023
National Laboratory of the Rockies Golden, CO

- Designed a test setup for switch characterization for power electronics.
- Performed detailed simulation, printed circuit board layout, and investigated advanced sensing techniques to achieve high-bandwidth performance.

Researcher in the Renewable and Appropriate Energy Lab July 2020 – Aug. 2020
University of California, Berkeley Berkeley, CA

- Extended research beginning in the SUPERB REU under Professor Daniel Kammen in the Energy Resources Group.
- Performed statistical and spatial analyses in Python and GIS to isolate candidate communities for microgrid development.

SUPERB Research Experience for Undergraduates June 2019 – Aug. 2019
University of California, Berkeley Berkeley, CA

- Analyzed geospatial data in R and GIS to classify community types in California with emphasis on environmental predictors.
- Selected as the sole student from the SUPERB REU to present research at NSF REU Research Symposium in Alexandria, VA, in October 2019.

Capstone to Work Research Team Sept. 2018 – May 2020
Smith College Northampton, MA

- Reviewed qualitative and quantitative data from a large pool of first year engineering employees to analyze the impacts of the engineering capstone experience on employment after college.

Publications

Refereed Journal Papers

[J7] S. Krishnan, **F. Giardine**, T. Mahbub, L. Horowitz, and R. C. N. Pilawa-Podgurski, "A High-Performance Flying Capacitor Multilevel Dual Active Bridge DC-DC Converter for Isolated Automotive Systems," *IEEE Open Journal of Power Electronics*, vol. 7, pp. 2701–2713, 2026. DOI: 10.1109/OJPEL.2026.3733626

[J6] **F. Giardine**, S. Krishnan, Y. Wu, L. Horowitz, and R. C. N. Pilawa-Podgurski, "A Variable Frequency Technique for EMI and Efficiency Improvements in High-Level Count Flying Capacitor Multilevel Converters," *IEEE Open Journal of Power Electronics*, pp. 267–280, 2025. DOI: 10.1109/OJPEL.2025.3646981

[J5] N. C. Brooks, **F. Giardine**, and R. C. N. Pilawa-Podgurski, "DC-Link Capacitors for Twice-Line Frequency Power Decoupling: Design-Oriented Figures-of-Merit With Empirical Application," *IEEE Transactions on Power Electronics*, vol. 39, no. 6, pp. 6569–6573, 2024. DOI: 10.1109/TPEL.2023.3305153

[J4] J. Ford, M. Paretti, D. Kotys-Schwartz, S. Howe, C. Gerwitz, J. Deters, T. Mahmud, R. Ott, N. Alvarez, D. Knight, C. Hernandez, L. Rosenbauer, A. Kary, and **F. Giardine**, "Transitioning from Capstone Design Courses to Workplaces: A Study of New Engineers' First Three Months," *International Journal of Engineering Education*, 2019.

Journal Papers (In Review/Revision)

[**J3**] **F. Giardine**, K. Fernandez, and R. C. N. Pilawa-Podgurski, "A Two-Stage Non-Isolated Hybrid Switched-Capacitor Microinverter Utilizing a Multi-Resonant DC-DC Stage with Startup Functionality," *IEEE Open Journal of Power Electronics*, [Conditionally Accepted].

[**J2**] **F. Giardine**, S. Krishnan, M. Nerenberg*, and R. C. N. Pilawa-Podgurski, "On the Scaling of Common-Mode EMI in the Flying Capacitor Multilevel Converter," [In Revision].

[**J1**] E. Krause, **F. Giardine**, N. Biesterfeld, and R. C. N. Pilawa-Podgurski, "Analysis of Steady-State Balancing in the Flying Capacitor Multilevel Converter Considering Capacitor Voltage Ripple," [Revised/Resubmitted].

Refereed Conference Papers

[**C11**] Y. Zhao*, J. Armstrong*, **F. Giardine**, and R. C. N. Pilawa-Podgurski, "A High-Performance 6-Level Flying Capacitor Multilevel Totem-Pole Bidirectional DC-AC Converter for Single-Phase Systems," in *2026 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2026, pp. 71–77. DOI: 10.1109/APEC51134.2026.11516908

[**C10**] S. Krishnan, **F. Giardine**, S. T. Mahbub, L. Horowitz, and R. C. N. Pilawa-Podgurski, "A High-Performance Flying Capacitor Multilevel Dual Active Bridge DC-DC Converter for Isolated Automotive Systems," in *2026 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2026, pp. 3621–3627. DOI: 10.1109/APEC51134.2026.11516761

[**C9**] **F. Giardine**, S. Krishnan, M. Nerenberg*, and R. C. N. Pilawa-Podgurski, "An Analytical Model for Common-Mode EMI in Flying Capacitor Multilevel Converters," in *2025 IEEE 26th Workshop on Control and Modeling for Power Electronics (COMPEL)*, 2025, pp. 1–7. DOI: 10.1109/COMPEL57166.2025.11121281

[**C8**] E. Krause, N. Biesterfeld, **F. Giardine**, and R. C. N. Pilawa-Podgurski, "Analysis of Steady-State Balancing in the Flying Capacitor Multilevel Converter Considering Capacitor Voltage Ripple," in *2025 IEEE 26th Workshop on Control and Modeling for Power Electronics (COMPEL)*, 2025, pp. 1–8. DOI: 10.1109/COMPEL57166.2025.11121246

[**C7**] S. Krishnan, **F. Giardine**, J. Zou, R. A. Abramson, and R. C. N. Pilawa-Podgurski, "Design and Implementation of a GaN-Based, Cascaded Isolated DC-DC Converter for Satellite Applications," in *2025 IEEE/AIAA Transportation Electrification Conference and Electric Aircraft Technologies Symposium (ITEC+EATS)*, 2025, pp. 1–6. DOI: 10.1109/ITEC63604.2025.11097988

[**C6**] **F. Giardine**, S. Krishnan, L. Horowitz, and R. C. N. Pilawa-Podgurski, "A Variable Frequency Technique for EMI and Efficiency Improvements in High-Level Count Flying Capacitor Multilevel Converters," in *2025 IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2025, pp. 151–156. DOI: 10.1109/APEC48143.2025.10977102

[**C5**] **F. Giardine**, Y. Wu, and R. C. N. Pilawa-Podgurski, "A Variable Switching Frequency Control Technique for DC-AC Flying Capacitor Multilevel Converters to Improve Efficiency and Inductor Utilization," in *2024 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2024, pp. 3473–3478. DOI: 10.1109/ECCE55643.2024.10861273

[**C4**] **F. Giardine**, K. Fernandez, and R. C. N. Pilawa-Podgurski, "A Two-Stage Non-Isolated Hybrid Switched-Capacitor Microinverter Utilizing a Fixed-Ratio Resonant DC-DC Stage with Startup Functionality and Flying Capacitor Multilevel DC-AC Stage," in *2024 Energy Conversion Congress & Expo Europe (ECCE Europe)*, [Best Paper Award], 2024, pp. 1–7. DOI: 10.1109/ECCEEurope62508.2024.10752033

[**C3**] **F. Giardine**, N. C. Brooks, K. Fernandez, and R. C. Pilawa-Podgurski, "Utilizing Harmonic Injection to Reduce Energy Storage and Required Capacitance in an Active Series-Stacked Energy Buffer for Single-Phase Systems," in *2022 IEEE 23rd Workshop on Control and Modeling for Power Electronics (COMPEL)*, 2022, pp. 1–7. DOI: 10.1109/COMPEL53829.2022.9829980

[**C2**] C. Gerwitz, **F. Giardine**, R. Ott, and A. Kary, "Women's Unique Challenges in the Transitions to Engineering Work," in *2020 ASEE Annual Conference & Exposition*, 2020. DOI: 10.18260/1-2--35588

[**C1**] S. Howe, M. Paretti, R. Ott, J. Deters, C. Gerwitz, **F. Giardine**, C. Hernandez, and A. Kary, "Women's Experiences in the Transition from Capstone Design Courses to Engineering Workplaces," in *2019 ASEE Annual Conference & Exposition*, 2019. DOI: 10.18260/1-2--33566

*Denotes undergraduate student

Teaching

University of California, Berkeley — Berkeley, CA

Graduate Student Instructor — Spring 2024, 2025
- EE 108: Introduction to Electric Power and Renewable Energy

Guest Lecturer
- EE 113B: Power Electronics Design – "Electromagnetic Interference" — April 7, 2026

Smith College — Northampton, MA

Engineering Master Tutor — Spring 2019, Fall 2019, Spring 2020
- Provided targeted course support in Engineering Circuit Theory (EGR 220), Engineering Mechanics (EGR 270), Thermodynamics (EGR 290), and Fluid Mechanics (EGR 374)

Quantitative Tutor — Fall 2018
- General campus support in STEM coursework (Calculus I-III, statistics)

Professional Development

Scholarship and Teaching for Engineering Postdocs program (UIUC) — Fall 2026 - Spring 2027

Certificate in Teaching and Learning in Higher Education (UC Berkeley) — Fall 2025

Mentoring

University of Illinois Urbana-Champaign — Urbana, IL

Jay Lee (Undergraduate) — Sept. 2026 – Present
- Ripple-Free Multilevel Power Conversion

Abhinav Srinivasan (Undergraduate) — Sept. 2026 – Present
- 800 V Power Conversion for Datacenters

Charlie Gunderson (Undergraduate) — Sept. 2026 – Present
- Coupled Inductors for High-Performance Converter Design

University of California, Berkeley — Berkeley, CA

Jack Armstrong (Undergraduate/Graduate, *now pursuing Ph.D at UC Berkeley*) — May 2025 – Present
- Control for a High-Performance Totem-Pole DC-AC/AC-DC Converter [C11]
- High Performance Active Capacitor for DC-Link Ripple Reduction

Stephanie Schaupp (Undergraduate) — Sept. 2025 – May 2026
- High Voltage Safety Enclosure Development

Yayun Zhao (Undergraduate, *now pursuing Ph.D at UT Austin*) — Oct. 2024 – March 2026
- Design of a High-Performance Totem-Pole DC-AC/AC-DC Converter [C11]

Nikhil Ograin (Undergraduate, *now at Apple*) — Sept. 2024 – May 2025
- A Resonant Flying Capacitor Multilevel Inverter

Service and Leadership

University of California, Berkeley — Berkeley, CA

IEEE Power and Energy Society Student Chapter — Jan. 2021 – Dec. 2025
- 2023: Chair
- 2022, 2024: Vice Chair
- 2021: Member-at-Large

Berkeley EECS Graduate Admissions Reader — 2023, 2024

EECS Graduate Student Association Committee Member — Sept. 2021 – Sept. 2022

Manuscript Reviewer — 2022 – Present
- IEEE Transactions on Power Electronics
- IEEE Journal of Emerging and Selected Topics in Industrial Electronics
- IEEE Applied Power Electronics Conference
- IEEE Energy Conversion Congress and Exposition